

ADDITIONAL PROBLEM SHEET 4: SOLUTIONS

1. Turing conditions (Exam section A type)

- (a) The homogeneous equilibrium satisfies

$$F(u, v) = 0, \quad G(u, v) = 0. \quad (1)$$

- (b) This homogeneous equilibrium is unstable to small variations in density. This instability develops into an inhomogeneous pattern in the form often of spots or stripes (this being a Cartesian grid). The linearised version of the main equations has solutions in the form

$$\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} e^{\lambda(k_s^{ij})t} \cos\left(\frac{i\pi}{L_1}x_1\right) \cos\left(\frac{j\pi}{L_2}x_2\right), \quad (2)$$

where

$$k_s^{ij} = \frac{i^2\pi^2}{L_1^2} + \frac{j^2\pi^2}{L_2^2} \quad (3)$$

is the pattern number. The function $\lambda(k_s)$ dictates if the mode decays exponentially $\lambda(k_s) < 0$, or grows, $\lambda(k_s) \geq 0$. The growing modes form the pattern.

The first two conditions,

$$F_u + G_v < 0, \quad F_u G_v - G_u F_v > 0, \quad (4)$$

ensure that the $k_s^{00} = 0$ mode, the homogeneous perturbation, always decays.

The other two conditions,

$$G_v + DF_u > 0, \quad (G_v + DF_u)^2 - 4D(F_u G_v - G_u F_v) > 0, \quad (5)$$

ensure there is some finite set of wave modes $k_s \in [k_s^{\min}, k_s^{\max}]$ for which $\lambda(k_s^{ij}) > 0$ and hence a pattern can form.

- (c) If $F_u > 0$ the first condition requires $G_v < 0$ and $|G_v| > |F_u|$. The third condition then shows

$$D|F_u| - |G_v| > 0 \implies D > \frac{|G_v|}{|F_u|}, \quad (6)$$

and hence $D > 1$.

2. Periodic boundary conditions (Exam section B type)

- (a) A homogeneous equilibrium is one for which both spatial and temporal derivatives vanish. In the case at hand the equations of a homogenous equilibrium (u_0, v_0) are

$$F(u_0, v_0) = G(u_0, v_0) = 0. \quad (7)$$

(b) To $\mathcal{O}(\varepsilon)$ the main equations are

$$\frac{\partial u_1}{\partial t} = \nabla^2 u_1 + \gamma (F_u u_1 + F_v v_1), \quad (8)$$

$$\frac{\partial v_1}{\partial t} = D \nabla^2 v_1 + \gamma (G_u u_1 + G_v v_1), \quad (9)$$

where, for example, $F_u = \partial F / \partial u$, evaluated at $u = u_0$. We seek solutions in the form $u_1 = \text{Re} (A e^{i\mathbf{k} \cdot \mathbf{x} + \lambda t})$ and $v_1 = \text{Re} (B e^{i\mathbf{k} \cdot \mathbf{x} + \lambda t})$; then our linearised equations take the form

$$\lambda \begin{pmatrix} u \\ v \end{pmatrix} = \mathbf{J} \begin{pmatrix} u \\ v \end{pmatrix}, \quad (10)$$

where

$$\mathbf{J} = \begin{pmatrix} -(k_1^2 + k_2^2) + \gamma F_u & \gamma F_v \\ \gamma G_u & -D(k_1^2 + k_2^2) + \gamma G_v \end{pmatrix}. \quad (11)$$

We now apply the boundary conditions. Consider the x_1 boundary conditions. Depending on the choice of the constants A (or B) the solutions will take the form

$$A e^{ik_2 x_2 + \lambda t} \sin(k_1 x_1) \quad \text{or} \quad A e^{ik_2 x_2 + \lambda t} \cos(k_1 x_1). \quad (12)$$

For the sine case the boundary conditions require

$$\sin(0) = \sin(k_1 L_1), \quad (13)$$

so $k_1 L_1 = n\pi$ for $n \in 0, 1, 2, 3, \dots$

For the cosine case we require

$$\cos(0) = \cos(k_1 L_1), \quad (14)$$

so $k_1 L_1 = n\pi$ for $n \in 0, 1, 2, 3, \dots$

One can easily apply this argument to the x_2 direction. The quoted solution is then the sum over all permissible combinations.

Finally, in order to solve eq. (10) for nontrivial solutions u_1 and v_1 we require $\det(\mathbf{J} - \lambda \mathbf{I}) = 0$ where $\mathbf{I}$ is the identity matrix. Solutions to this take the form

$$\lambda(i, j) = \frac{1}{2} \left[\text{tr } \mathbf{J} \pm \sqrt{(\text{tr } \mathbf{J})^2 - 4 \det \mathbf{J}} \right] \quad (15)$$

with

$$\text{tr } \mathbf{J} = \gamma(F_u + G_v) - k_s(i, j)(D + 1), \quad (16)$$

$$\det \mathbf{J} = D k_s(i, j)^2 - \gamma k_s(i, j) (D F_u + G_v) + \gamma^2 (F_u G_v - F_v G_u), \quad (17)$$

and

$$k_s(i, j) = \pi^2 \left(\frac{i^2}{L_1^2} + \frac{j^2}{L_2^2} \right), \quad (18)$$

$$k_s(i, j) = \pi^2 \left(\frac{i^2}{L_1^2} + \frac{4j^2}{L_2^2} \right), \quad (19)$$

$$k_s(i, j) = \pi^2 \left(\frac{4i^2}{L_1^2} + \frac{j^2}{L_2^2} \right), \quad (20)$$

$$k_s(i, j) = \pi^2 \left(\frac{4i^2}{L_1^2} + \frac{4j^2}{L_2^2} \right), \quad (21)$$

depending on the particular solution type.

(c) In this case the linearised solutions u_1 and v_1 must satisfy

$$u_1(x_1, 0) = u_1(x_1, L_2) = u_1(0, x_2) = u_1(L_1, x_2) = 0, \quad (22)$$

$$v_1(x_1, 0) = v_1(x_1, L_2) = v_1(0, x_2) = v_1(L_1, x_2) = 0. \quad (23)$$

In this case the cosine form of the solutions cannot satisfy the boundary conditions as $\cos(0) = 1$. The sine boundary condition then require

$$k_1 = \frac{n\pi}{L_1} \quad \text{and} \quad k_2 = \frac{m\pi}{L_2}, \quad (24)$$

so that

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} e^{\lambda(i,j)t} C_1^{ij} \sin\left(\frac{i\pi}{L_1}x_1\right) \sin\left(\frac{j\pi}{L_2}x_2\right). \quad (25)$$

The summation accounts for the fact that the zeroth mode $i = j = 0$ is 0. The form of $\lambda(i, j)$ will be as in the periodic case with

$$k_s(i, j) = \pi^2 \left(\frac{i^2}{L_1^2} + \frac{j^2}{L_2^2} \right). \quad (26)$$

(d) These two conditions enforce that constant changes in the system, the $i = j = 0$ modes, do not grow in time. This is necessary to ensure any potential patterns are not drowned out. In the Dirichlet case (from the previous part) there is no non-zero $i = j = 0$ mode, so we have no need to enforce the conditions. For the periodic case the $\cos() \cos()$ solution is potentially non-zero (if $C_4^{00} \neq 0$) so we must enforce these two conditions.

3. Nonlinear diffusion (Exam section B type)

(a) Homogeneous equilibria satisfy

$$F(u, v) = 0, \quad G(u, v) = 0. \quad (27)$$

(b) The boundary conditions are

$$\frac{\partial u}{\partial x_1}(0, x_2) = \frac{\partial u}{\partial x_1}(L_1, x_2) = \frac{\partial u}{\partial x_2}(x_1, 0) = \frac{\partial u}{\partial x_2}(x_1, L_2) = 0, \quad (28)$$

$$\frac{\partial v}{\partial x_1}(0, x_2) = \frac{\partial v}{\partial x_1}(L_1, x_2) = \frac{\partial v}{\partial x_2}(x_1, 0) = \frac{\partial v}{\partial x_2}(x_1, L_2) = 0. \quad (29)$$

I would accept other notation as long as it is clear, e.g.

$$\left. \frac{\partial u}{\partial x_1} \right|_{x_1=0} = 0, \quad \forall x_2 \in [0, L_2]. \quad (30)$$

(c) We assume the homogeneous equilibria are labelled u_0 and v_0 . We assume $u(x, t) = u_0 + \varepsilon u_1(x, t)$ and $v(x, t) = v_0 + \varepsilon v_1(x, t)$. We expand the main equations; the linear and F, G expansions are clear and obvious. For the nonlinear terms we have

$$\nabla(u^2) = \nabla(u_0 + \varepsilon 2u_0 u_1) + \mathcal{O}(\varepsilon^2) = \varepsilon 2u_0 \nabla u_1 + \mathcal{O}(\varepsilon^2), \quad (31)$$

$$\nabla v \cdot \nabla(u^2) = \mathcal{O}(\varepsilon^2), \quad (32)$$

$$\nabla^2 u = \varepsilon \nabla^2 u_1 \implies v \nabla^2 u = \varepsilon v_0 \nabla^2 u_1 + \mathcal{O}(\varepsilon^2). \quad (33)$$

So to $\mathcal{O}(\varepsilon)$ our equations read

$$\frac{\partial u_1}{\partial t} = \nabla^2 u_1 + \gamma (F_u u_1 + F_v v_1), \quad (34)$$

$$\frac{\partial v_1}{\partial t} = D \nabla^2 v_1 + \gamma (G_u u_1 + G_v v_1) + v_0 \nabla^2 u_1, \quad (35)$$

where $F_u = \partial F / \partial u$ evaluated at $u = u_0$, etc.

We seek solutions in the form $u_1 = \text{Re} (A e^{i \mathbf{k} \cdot \mathbf{x} + \lambda t})$ and $v_1 = \text{Re} (B e^{i \mathbf{k} \cdot \mathbf{x} + \lambda t})$. Our linearised equations then take the form

$$\lambda \begin{pmatrix} u \\ v \end{pmatrix} = \mathbf{J} \begin{pmatrix} u \\ v \end{pmatrix}, \quad (36)$$

where

$$\mathbf{J} = \begin{pmatrix} -k_s + \gamma F_u & \gamma F_v \\ -k_s v_0 + \gamma G_u & -D k_s + \gamma G_v \end{pmatrix} \quad (37)$$

and where $k_s = k_1^2 + k_2^2$.

In order to have nontrivial u_1, v_1 we require $\det(\mathbf{J} - \lambda \mathbf{I}) = 0$ so

$$\lambda(i, j) = \frac{1}{2} \left[\text{tr } \mathbf{J} \pm \sqrt{(\text{tr } \mathbf{J})^2 - 4 \det \mathbf{J}} \right] \quad (38)$$

with

$$\text{tr } \mathbf{J} = \gamma (F_u + G_v) - k_s (D + 1), \quad (39)$$

$$\det \mathbf{J} = D k_s^2 - \gamma k_s (D F_u + G_v - v_0 F_v) + \gamma^2 (F_u G_v - F_v G_u). \quad (40)$$

The boundary conditions require

$$k_s = \pi^2 \left(\frac{n_1^2}{L_1^2} + \frac{n_2^2}{L_2^2} \right). \quad (41)$$

First we require the $k_s = 0$ mode be asymptotically stable (decay) so we require $\text{tr } \mathbf{J} < 0$ and $\det \mathbf{J} > 0$ when $k_s = 0$, i.e.

$$F_u + G_v < 0, \text{ and } F_u G_v - F_v G_u > 0. \quad (42)$$

Next we want some $k_s > 0$ such that some modes (n_1, n_2) grow. Since $(F_u + G_v) < 0$ the trace must be negative, so this can only occur if $\det \mathbf{J} < 0$. Since $F_u G_v - F_v G_u > 0$ we require

$$D F_u + G_v - v_0 F_v > 0, \quad (43)$$

a necessary but not sufficient condition.

Finally, since the quadratic component of k_s is positive we can guarantee solutions if the quadratic $\det \mathbf{J}$ crosses the real axis, thus the discriminant of the quadratic $\det \mathbf{J}$ must be greater than 0, i.e.

$$(D F_u + G_v - v_0 F_v)^2 - 4 D (F_u G_v - F_v G_u) > 0. \quad (44)$$

- (d) If we set $u = 0$ we get $v = 0, 1$. If we set $v = 0$ we get $u = 0, 1$. So our equilibria are $(0, 0)$, $(1, 0)$ and $(0, 1)$.

(e) We have

$$F_u = 1 - 2u_0 - v_0, \quad F_v = -u_0, \quad G_v = 1 - 2v_0 + \gamma u_0, \quad G_u = \gamma v_0. \quad (45)$$

Consider the $(0, 0)$ equilibrium: We try the first condition

$$F_u + G_v = 2 > 0 \quad (46)$$

so we cannot satisfy the first condition.

Consider the $(1, 0)$ equilibrium: The first condition reads

$$F_u + G_v = \gamma < 0, \quad (47)$$

so we require that γ be negative. The second condition is

$$F_u G_v - F_v G_u = -(1 + \gamma) > 0 \quad (48)$$

so this requires that γ is less than -1 . The third condition reads

$$DF_u + G_v = -D + (1 + \gamma) > 0 \implies \gamma > D - 1. \quad (49)$$

We see that it is not possible to satisfy $\gamma < -1$ and $\gamma > D - 1$. So we cannot satisfy all the Turing conditions.

Now consider the $(0, 1)$ equilibrium:

$$F_u + G_v = -1 < 0. \quad (50)$$

The second condition requires

$$F_u G_v - F_v G_u = 0 > 0 \quad (51)$$

which cannot be satisfied.

So in fact we cannot have any pattern formation from homogeneous equilibria for which one of the densities is zero.

4. Patterns and polar coordinates (Exam section A, 2017)

(a) The homogeneous equilibrium satisfies

$$F(u, v) = 0, \quad G(u, v) = 0. \quad (52)$$

(b) This homogeneous equilibrium is unstable to small variations in density. This instability develops into an inhomogeneous pattern in the form of spots or stripes (this being a Cartesian grid). The linearised version of the main equations has solutions in the form

$$\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} e^{\lambda(k_s^{ij})t} \cos\left(\frac{i\pi}{L_1}x_1\right) \cos\left(\frac{j\pi}{L_2}x_2\right), \quad (53)$$

where

$$k_s^{ij} = \frac{i^2\pi^2}{L_1^2} + \frac{j^2\pi^2}{L_2^2}. \quad (54)$$

The function $\lambda(k_s)$ dictates if the mode decays exponentially $\lambda(k_s) < 0$, or grows, $\lambda(k_s) \geq 0$. The growing modes form the pattern.

The first two conditions, $F_u + G_v < 0$ and $F_u G_v - G_u F_v > 0$, ensure that the $k_s^{00} = 0$ mode, the homogeneous perturbation, always decays.

The other two conditions, $G_v + D F_u > 0$ and $(G_v + D F_u)^2 - 4D(F_u G_v - G_u F_v) > 0$, ensure there is some finite set of wave modes $k_s \in [k_s^{\min}, k_s^{\max}]$ for which $\lambda(k_s^{ij}) > 0$ and hence a pattern can form.

- (c) The patterns (solutions to the linearised system) take the form

$$\sum_{n=0}^{\infty} \sum_{i=1}^{\infty} \left\{ \left[A_{ni} J_n(\sqrt{k_{sn}^i} r) + B_{ni} Y_n(\sqrt{k_{sn}^i} r) \right] \cos(n\theta) + \left[C_{ni} J_n(\sqrt{k_{sn}^i} r) + E_{ni} Y_n(\sqrt{k_{sn}^i} r) \right] \sin(n\theta) \right\}. \quad (55)$$

Here J_n and Y_n are the Bessel functions of the first and second kind respectively, A_{ni}, B_{ni}, C_{ni} and E_{ni} are constants and k_{sn}^i is the i^{th} pattern number associated with the n^{th} Fourier mode.

5. More Turing analysis (Exam section B, 2017)

- (a) The boundary conditions are $\nabla u \cdot \hat{\mathbf{n}} = 0$ and $\nabla v \cdot \hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is the unit normal to ∂V .
- (b) The homogeneous equilibria are determined by $F(u_0, v_0) = 0$ and $G(u_0, v_0) = 0$.
- (c) We linearise our system by Taylor expanding $u \approx u_0 + \varepsilon u_1$ and $v \approx v_0 + \varepsilon v_1$. Denoting $F_u = \partial F / \partial u|_{u=u_0, v=v_0}$ etc. we have, at order ε ,

$$\begin{aligned} \frac{\partial u_1}{\partial t} &= \nabla^2 u_1 + \gamma F_u u_1 + \gamma F_v v_1 + u_0 \nabla^2 v_1 + v_0 \nabla^2 u_1, \\ \frac{\partial v_1}{\partial t} &= D \nabla^2 v_1 + \gamma G_u u_1 + \gamma G_v v_1. \end{aligned}$$

Assuming $u_1 = C_u e^{i\mathbf{k} \cdot \mathbf{x} + \lambda t}$ and $v_1 = C_v e^{i\mathbf{k} \cdot \mathbf{x} + \lambda t}$. We can write this as a matrix equation

$$\lambda \begin{pmatrix} u_1 \\ v_1 \end{pmatrix} = \mathbf{J} \begin{pmatrix} u_1 \\ v_1 \end{pmatrix}, \quad (56)$$

where

$$\mathbf{J} = \begin{pmatrix} -k_s(1 + v_0) + \gamma F_u & -k_s u_0 + \gamma F_v \\ \gamma G_u & -D k_s + \gamma G_v \end{pmatrix} \quad (57)$$

and $k_s = \mathbf{k} \cdot \mathbf{k}$. So solutions can only occur when $\det(\mathbf{J} - \lambda \mathbf{I}) = 0$. We know from the notes that solutions take the form

$$\lambda = \frac{1}{2} \left[\text{tr } \mathbf{J} \pm \sqrt{(\text{tr } \mathbf{J})^2 - 4 \det \mathbf{J}} \right], \quad (58)$$

where

$$\begin{aligned} \text{tr } \mathbf{J} &= \gamma(F_u + G_v) - k_s(D + v_0 + 1), \\ \det \mathbf{J} &= D(1 + v_0)k_s^2 - k_s \gamma(G_v(1 + v_0) + D F_u - G_u u_0) + \gamma^2(F_u G_v - F_v G_u). \end{aligned} \quad (59)$$

We first need the homogeneous perturbations $k_s = 0$ to decay, thus we require $\text{tr} < 0$ and $\det > 0$, so

$$F_u + G_v < 0 \text{ and } F_u G_v - F_v G_u > 0. \quad (60)$$

This gives the first two conditions.

Next we want there to be some $k_s > 0$ for which $\lambda(k_s) > 0$, so they grow. Since $F_u + G_v < 0$ the trace must be negative (v_0 must be positive) so we can only have positive eigenvalues λ when $\det < 0$, i.e.

$$D(1 + v_0)k_s^2 - k_s\gamma(G_v(1 + v_0) + DF_u - G_u u_0) + \gamma^2(F_u G_v - F_v G_u) > 0. \quad (61)$$

Since $F_u G_v - F_v G_u > 0$ and $D(1 + v_0) > 0$ this can only occur if

$$G_v(1 + v_0) + DF_u - G_u u_0 > 0, \quad (62)$$

the third condition. But this is necessary not sufficient, finally, if we want real k_s to solve $\det \mathbf{J} = 0$ we need the discriminant of the quadratic in k_s to be positive, i.e.

$$(G_v(1 + v_0) + DF_u - G_u u_0)^2 - 4D(1 + v_0)(F_u G_v - F_v G_u) \geq 0, \quad (63)$$

the final condition.

(d) First $F = 0$ when $u = 0$, so $v = 1$. We also solve $F = 0$ by setting

$$u = \frac{k_1 v}{k_2 + v}. \quad (64)$$

So either $v = 1$ or $u = 1$ and eq. (64) gives the final two equilibria. The three equilibria (u_0, v_0) are

$$(0, 1), \quad \left(1, \frac{k_2}{k_1 - 1}\right), \quad \left(\frac{k_1}{1 + k_2}, 1\right). \quad (65)$$

We have

$$\begin{aligned} F_u &= \frac{k_1 v_0}{k_2 + v_0} - 2u_0, & F_v &= u_0 \left(\frac{k_1}{k_2 + v_0} - \frac{k_1 v_0}{(k_2 + v_0)^2} \right), \\ G_u &= v_0 - 1, & G_v &= u_0 - 1. \end{aligned} \quad (66)$$

First the case $u_0 = 0, v_0 = 1$:

$$F_u G_v - F_v G_u = -\frac{k_1}{k_2 + 1} > 0. \quad (67)$$

This is a contradiction as $k_1, k_2 > 0$, so this equilibrium cannot satisfy eq. (59).

Note for the next two cases, $u_0 = \frac{k_1 v_0}{k_2 + v_0}$, so

$$F_u = \frac{k_1 v_0}{k_2 + v_0} - 2u_0 = -u_0. \quad (68)$$

Next $(1, \frac{k_2}{k_1 - 1})$: The third condition gives

$$G_v(1 + v_0) + DF_u - u_0 G_u = -D - (v_0 - 1) > 0. \quad (69)$$

So at the very least we need $v_0 < 1$. Now the second condition

$$F_u G_v - F_v G_u = -F_v G_u > 0. \quad (70)$$

If $v_0 < 1$ then $G_u < 0$ and we see that

$$F_v = \frac{u_0 k_1}{k_2 + v_0} \left(1 - \frac{v_0}{k_2 + v_0} \right). \quad (71)$$

must be positive. So the second condition is satisfied if $v_0 < 1$. Finally the first condition

$$F_u + G_v = -u_0 < 0. \quad (72)$$

is satisfied. So if $v_0 < 1 - D$, then the first three conditions will be satisfied.

The final case is $(\frac{k_1}{k_2+1}, 1)$. The second condition gives

$$F_u G_v - F_v G_u = u_0(1 - u_0) > 0, \quad (73)$$

so we require $u_0 < 1$. The third condition is

$$G_v(1 + v_0) + D F_u - u_0 G_u = (u_0 - 1)(1 + v_0) - D u_0 > 0. \quad (74)$$

But we needed $u_0 < 1$ so we cannot satisfy this condition.

6. Boundary condition effects (Exam section A type)

- (a) The homogeneous equilibrium defined by $F(u, v) = 0$, $G(u, v) = 0$ is unstable to small variations in density. This instability develops into an inhomogeneous pattern in the form of spots or stripes (this being a Cartesian grid). The linearised version of the main equations has solutions of the form

$$\sum_{i=0}^n \sum_{j=0}^m e^{\lambda(k_s^{ij})t} \cos\left(\frac{n_i \pi}{L_1} x_1\right) \cos\left(\frac{m_j \pi}{L_2} x_2\right), \quad (75)$$

where

$$k_s^{ij} = \frac{n_i^2 \pi^2}{L_1^2} + \frac{m_j^2 \pi^2}{L_2^2}. \quad (76)$$

The function $\lambda(k_s)$ dictates if the mode decays exponentially, $\lambda(k_s) < 0$, or grows, $\lambda(k_s) \geq 0$. The growing modes form the pattern.

The first two conditions, $F_u + G_v < 0$ and $F_u G_v - G_u F_v > 0$, ensure that the $k_s^{00} = 0$ mode, the homogeneous perturbation, always decays. The other two conditions $G_v + D F_u > 0$ and $(G_v + D F_u)^2 - 4D(F_u G_v - G_u F_v) > 0$, ensure there is some finite set of wave modes, $k_s \in [k_s^{\min}, k_s^{\max}]$, for which $\lambda(k_s^{ij}) > 0$, and hence a pattern can form.

- (b) In the case of periodic boundary conditions both the $\sin(k_{1,2}x)$ and $\cos(k_{1,2}x)$ spatial functions are permissible solutions. This means the $n = m = 0$ ($k_s = 0$) homogeneous case is permissible and the two stated conditions are necessary. In the second case the fixed-value boundary conditions at $x_1 = 0, x_1 = L$ means we can have no $n_1 = 0$ mode, thus no $k_s = 0$ pattern. This means the two conditions are unnecessary.

7. Enzyme-driven Turing patterns (Exam section A type)

(a) The first condition gives

$$1 - \frac{2a}{u_0} - u_0^2 < 0 \implies a > \frac{u_0(1 - u_0^2)}{2}. \quad (77)$$

The second condition is trivially satisfied,

$$F_u G_v - F_v G_u = u_0^2 > 0, \quad (78)$$

so no condition on a .

The third condition gives

$$D(1 - \frac{2a}{u_0}) - u_0^2 \implies a < \frac{u_0}{2} \left(1 - \frac{u_0^2}{D}\right). \quad (79)$$

(b) As

$$\frac{u_0}{2} \left(1 - \frac{2u_0}{\sqrt{D}} - \frac{u_0^2}{D}\right) < \frac{u_0}{2} \left(1 - \frac{u_0^2}{D}\right), \quad (80)$$

we see that eq. (79) is satisfied if

$$a < \frac{u_0}{2} \left(1 - \frac{2u_0}{\sqrt{D}} - \frac{u_0^2}{D}\right). \quad (81)$$

Next,

$$\frac{u_0}{2} \left(1 + \frac{2u_0}{\sqrt{D}} - \frac{u_0^2}{D}\right) > \frac{u_0}{2} (1 - u_0^2), \quad (82)$$

so satisfying the other condition 4 inequality,

$$a > \frac{u_0}{2} \left(1 + \frac{2u_0}{\sqrt{D}} - \frac{u_0^2}{D}\right), \quad (83)$$

ensures eq. (77) is satisfied. But we must check *each* condition 4 inequality satisfies *both* conditions 1 and 3.

We also note that

$$\frac{u_0}{2} \left(1 + \frac{2u_0}{\sqrt{D}} - \frac{u_0^2}{D}\right) > \frac{u_0}{2} \left(1 - \frac{u_0^2}{D}\right) > \frac{u_0}{2} \left(1 - \frac{2u_0}{\sqrt{D}} - \frac{u_0^2}{D}\right). \quad (84)$$

So it is not possible to satisfy

$$a > \frac{u_0}{2} \left(1 + \frac{2u_0}{\sqrt{D}} - \frac{u_0^2}{D}\right). \quad (85)$$

and

$$a < \frac{u_0}{2} \left(1 - \frac{u_0^2}{D}\right). \quad (86)$$

That is to say one of the possible cases for condition 4 holding would fail to also satisfy condition 3. We are left with only one possibly viable inequality to satisfy condition 4,

$$a < \frac{u_0}{2} \left(1 - \frac{2u_0}{\sqrt{D}} - \frac{u_0^2}{D}\right), \quad (87)$$

which we know from above satisfies condition 3. We need to check it can satisfy condition 1.

Indeed we can see it is possible that

$$\frac{u_0}{2} (1 - u_0^2) < \frac{u_0}{2} \left(1 - \frac{2u_0}{\sqrt{D}} - \frac{u_0^2}{D} \right). \quad (88)$$

in the limit $D \rightarrow \infty$.